

Mahmoud Hosny

Software Engineer · ML & AI · Data Analysis

✉ mahmoudandmohamed391@gmail.com 🔗 linkedin.com/in/mahmoud-hosny-46627332b 📄 github.com/7oda7osny

PROFILE

Software engineer with hands-on experience across the full development spectrum — from object-oriented system design in Java and C++ to building and evaluating machine learning models in Python, R, and SAS. Comfortable with the complete ML pipeline: data wrangling, feature engineering, model training, cross-validation, and evaluation. Passionate about building things that are clean, correct, and useful.

TECHNICAL SKILLS

Languages	Java, Python, C++, R, SQL, SAS, JavaScript, HTML, CSS3, SQLite
ML & Data	Linear Regression, KNN, Naïve Bayes, Classification, Gradient Descent, Cross Validation, Feature Engineering, Data Mining, EDA
Tools	R Studio, Java Swing, OOP Design, A* Search, Git

PROJECTS

3×3 Puzzle Game with AI Solver — *Python · A* Algorithm*

- Built an interactive sliding puzzle game with an AI solver using the A* search algorithm, finding optimal solutions via heuristic pathfinding.

Library Administration System — *Java · OOP*

- Designed a full-featured library management system handling books, members, borrowing, and returns using clean object-oriented principles.

E-Commerce Website Backend — *Java · OOP*

- Built the backend for an e-commerce platform covering product listings, order management, and user accounts with solid OOP architecture.

City Supermarkets Analysis — *R Studio · Statistics · EDA*

- Performed exploratory and statistical analysis on supermarket data to uncover sales patterns, customer behavior, and regional trends.

Airplane BnB — Desktop App — *Java · Java Swing · SQLite*

- Developed a desktop booking application for airplane seats with a Swing GUI and SQLite persistence for users and reservations.

University Registration System — *Java · OOP*

- Built an academic registration system managing student enrollment, course scheduling, and grade tracking with a layered OOP design.

Hotel Page — Frontend — *HTML · CSS3 · JavaScript*

- Designed and built a responsive hotel landing page with CSS3 animations and vanilla JavaScript for interactive UI components.

E-Commerce Churn Prediction — *SAS · Classification*

- Predicted customer churn for an e-commerce platform using SAS classification models, identifying at-risk customers through data preprocessing and logistic modeling.

Student Grade Prediction — *Data Mining · Classification*

- Applied data mining techniques to predict student academic performance, including feature selection and evaluation of multiple classification models.

Car Price Prediction — *Python · ML*

- Compared Linear Regression, KNN, and Naïve Bayes for predicting used car prices, with cross-validation and metric-based model selection.

FIFA Players Performance Analysis — *Python · ML*

- Built regression and classification models on the FIFA players dataset — including player value prediction and performance tier classification, with feature engineering and cross-validation.

California House Pricing — *Python · Regression · Statistics*

- Comprehensive housing price regression study approached from statistical, gradient descent, and numerical perspectives with comparative model evaluation and visualizations.

E D U C A T I O N

Bachelor of Science — Computer Science / Software Engineering · *Alexandria Universty*

2024 — 2028